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	<u>(Detector Signals Input Test on NS3a/d Controllers)</u>	Page 1 of 4
		Products: NS3a/d Dimension Systems
Revision	Description	Date
B	Initial Release – (John Tedesco)	(2/5/10)

Subject: (Detector Signals Input Test on NS3a/d Controllers)

System(s) Affected: (NS3a/d Dimension Systems)

Action Category: (As Required)

Summary: (Testing the detector signals and eliminating the back panel as a possible source of detector signal problems.)

1. Remove any Extender box from line between the NS3a and Dimension Microscope. Make sure that the Dimension microscope is directly connected to the Front of the NS3a/d controller.
2. Change the SW settings to eliminate the extender.
 - a. For v6 go to tools→select microscope→edit and change extender to none. Click OK until realtime starts
 - b. For v5 go to the di logo in the top left corner of the left hand screen →microscope select→edit and change extender to none. Click Ok until realtime starts
3. Open the top of the NS3a controller.

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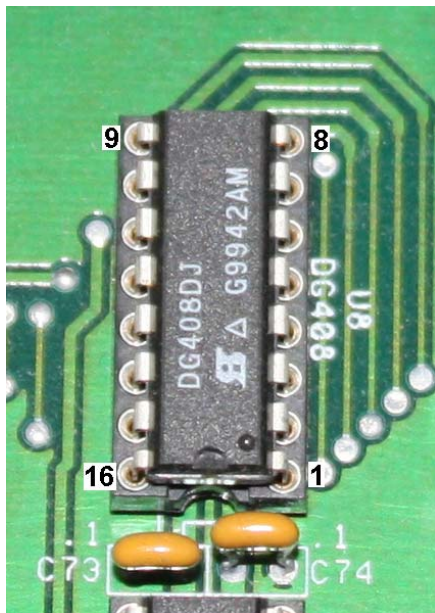


4. Locate the AUX board in the rear left corner of the controller box.



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5. Locate Chip U8 (near the middle of the board). Below is the pin designation for the chip



6. When the SW is started up and in realtime with no extender selected and no extender on the system the pins will be laid out as the following:

Pin4 – Lateral (Horizontal) Deflection

Pin5 – Vertical Deflection

Pin6 – Sum Signal

7. Place your laser on your cantilever and center the spot in the oval viewing window on your scanner

BE VERY CAREFUL ON THE NEXT PORTION OF THIS TEST. IF YOU ACCIDENTALLY TOUCH TWO PINS TOGETHER AT THE SAME TIME THIS COULD DAMAGE THE NANOSCOPE CONTROLLER. ONLY ATTEMPT THIS TEST IF A VEECO REPRESENTATIVE HAS ASKED YOU TO DO SO.



8. Use a voltmeter or oscilloscope (make sure to set for DC voltage) to test each pin. Use the controller chassis as the reference ground (touch the black lead of the voltmeter to any metal piece on the controller box that is not the power supply or a PC board).

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9. Start with Pin6 and measure the Sum. It should be a negative voltage. For tapping mode tips with no aluminum coating this value should be near -2V, for aluminum coated tips near -4V, for SiN gold coated tips near -6V.
10. Next test the vertical deflection pin (pin5). When testing use the vertical deflection adjust knob (pic in appendix A) on the left side of the scanner. When rotating this knob it should be possible to get this pin to go to +/-12V. for example, hold the voltmeter's red lead to pin5, hold the black lead to the controller chassis and then rotate the upper left knob of the dimension scanner all at the same time (yes, you will probably need 2 people). The values should change and as the knob is rotated counter-clockwise the value should move more negative, clockwise more positive.
11. Do the same thing as in #11 but this time on pin 4 and rotate the bottom left knob of the scanner to adjust horizontal deflection (see appendix A for location of knob).
12. If all values come through properly on the test pins yet the detector screen is not reading them properly then the problem is in the controller. If the values do not come through as expected then the scanner or the electronics in the Dimension are to blame.

Appendix A:

